2024-1 Pattern Recognition

Project Report

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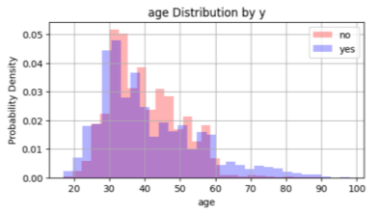
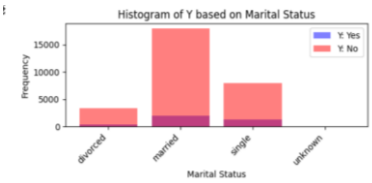
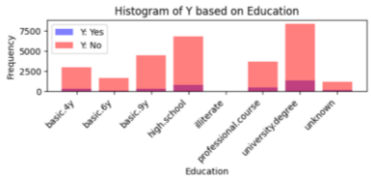
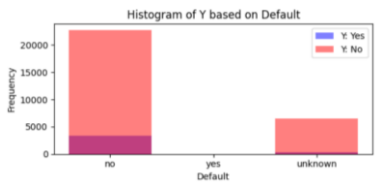
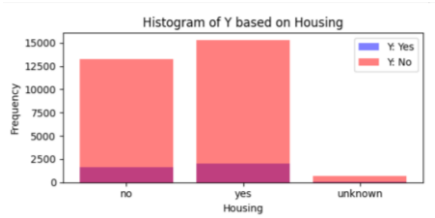
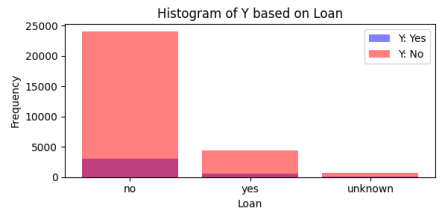
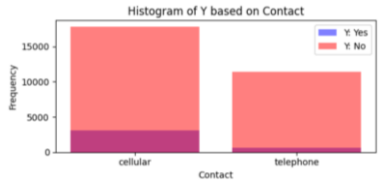
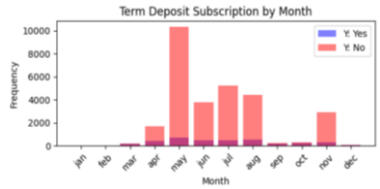
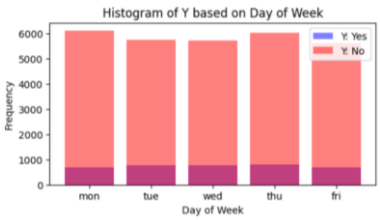
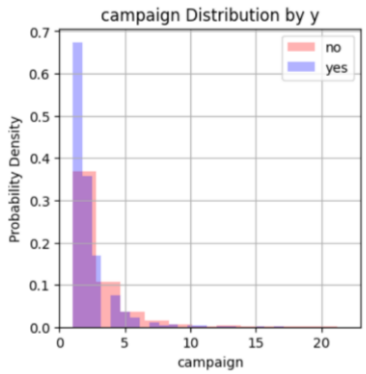
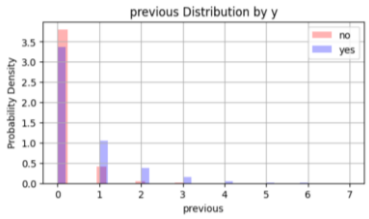
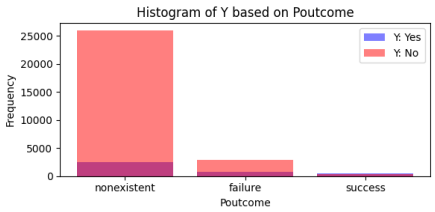
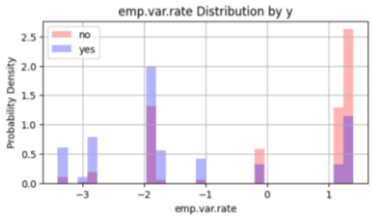
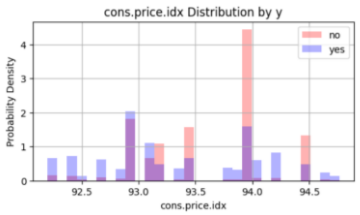
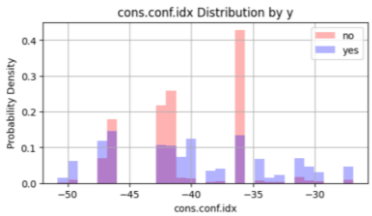
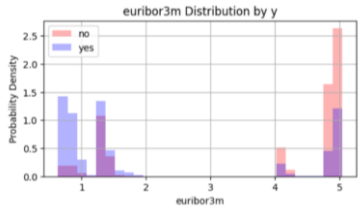
The goal of this project is to predict the subscription to a term deposit in the test set based on the analysis of the training set. In the training set, we will examine whether the output variable, 'y', is 'yes' or 'no' for each input variable (both categorical and continuous). This analysis aims to understand how each input variable relates to the desired target variable in the training set, providing insights into potential predictive features for the test set.

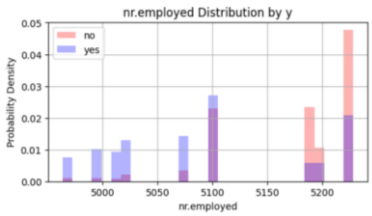
1. Data Exploration (with Pandas)

Examine the distribution of the categorical, continuous variable grouped by the target class.

	**2-age**: For the people above the age of 61, the percentage of Y(yes)s is much greater than N(no)s. Therefore, it is reasonable to regard people above 61 as Ys.
	3-job: Cannot find apparent features. Have to compare the jobs by income with our own background knowledge.
	**4-marital status**: Cannot find apparent features. Have to find the percentage of Ys among each category.
	**5-education**: Cannot find apparent features. Have to find the percentage of Ys among each category.

	**6-default**: Have to preprocess unknown values using methods such as substituting with mode values.
	**7-housing**: Have to preprocess unknown values using methods such as substituting with mode values.
	**8-loan**: Have to preprocess the unknown values using methods such as substituting with mode values.
	**9-contact**: Does not have a direct effect on the result.
	**10-month**: Does not have a direct effect on the result, but the value of category May is significantly greater than the others.
	**11-day of week**: Does not have a direct effect on the result.
	**12-campaign**: Cannot find apparent features. Have to find the percentage of Y(yes)s among each category.

	13-pdays: Should come out with a good strategy dealing with values containing 999.
	**14-previous**: As the number of previous contacts increase, it is improbable for the client to subscribe to the bank term deposit.
	**15-poutcome**: It seems optimal to ignore this variable since the percentage of the nonexistent values is overwhelming the other values.
	**16-emp.var.rate**: As the employment variation rate values are smaller, the percentage of Ys is larger.
	**17-cons.price.idx**: As the consumer price index values are smaller, the percentage of Ys is larger.
	**18-cons.conf.idx**: We can see that the consumer confidence index scores well for the people above 35, resulting in more subscriptions.
	**19-euribor3m** : Setting 3 as the standard, if it is greater than 3, it is more likely that the client will not subscribe to the deposit.

	**20-nr.employed**: Setting 5150 as the standard, if it is greater than 5150, it is more likely that the client will not subscribe to the deposit.

2. Data Preprocessing

1) Variable Selection: eliminate variables that are not utilized as inputs or that contain significant number of unknown or missing values

We dropped the variables that are unlikely to affect the result. We will decide whether or not to drop the ‘poutcome’ and ‘month’ variables based on the accuracy, f1, and ROC AUC scores in the Training & Validation stage.

Since the 'id' and 'contact' columns are not used, we used the drop() function to delete the columns, which is known as ‘complete case analysis’.

In addition, the 'day_of_week' column is also treated with a drop() function because there was no significant difference in the number of values.

```
df.drop("id", axis=1, inplace=True)
df.drop("contact", axis=1, inplace=True)
df.drop("day_of_week", axis=1, inplace=True)
```

2) Dealing with missing values

Among the missing value imputation methods, we tried to apply SimpleImputer and IterativeImputer to find the optimal performance. Generally, if there are more than 10% of missing values in the feature, we use the IterativeImputer and if there are less than 10%, we use the SimpleImputer.

SimpleImputer uses a method of replacing the missing value with the most frequent value, so we used the mode() function to process the missing value.

We used the mode() function, which processes missing values of categorical variables, to process unknown values in columns 'job', 'marital', 'education', 'default', 'housing', and 'loan', and replaced the unknown values to mode values. We also changed the original value itself by setting the inplace attribute as True.

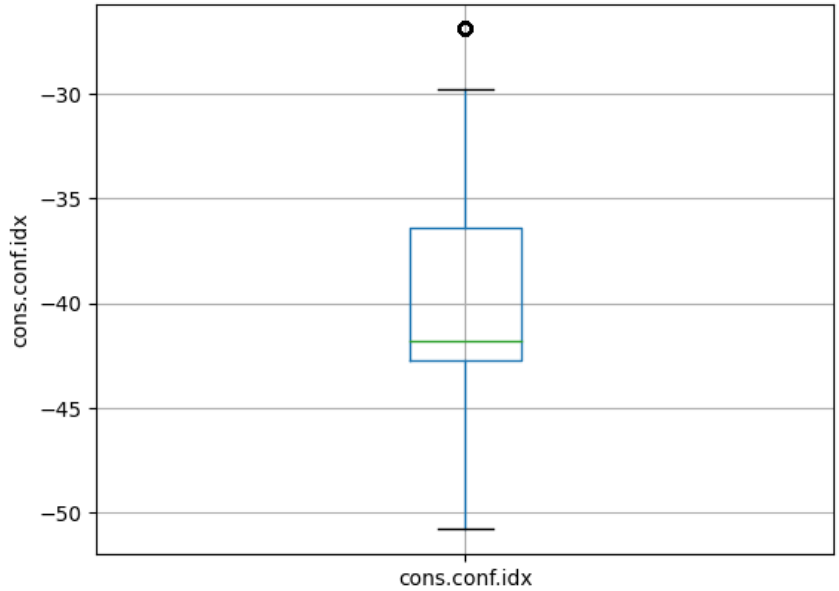
```
df.replace({'job': {'unknown': df["job"].mode()[0]}}, inplace = True)
df.replace({'marital': {'unknown': df["marital"].mode()[0]}}, inplace = True)
df.replace({'housing': {'unknown': df["housing"].mode()[0]}}, inplace = True)
df.replace({'loan': {'unknown': df["loan"].mode()[0]}}, inplace = True)
df
```

The ‘default’ variable had 6853 unknown values, which takes up about 20% of the total values. Since there were only two values 0 and 1 for the ‘default’ variable, just processing the ‘default’ variable with the IterativeImputer caused an error. The variable with the next highest unknown values was ‘education’ consisting about 4%. So, we decided to process the ‘education’ variable also with the IterativeImputer. To use the IterativeImputer, we first applied label encoding to ‘default’ and ‘education’ variables and replaced ‘unknown’ values to NaN. Below shows the imputation process using the IterativeImputer.

```
# IterativeImputer
imputer = IterativeImputer(max_iter=10, random_state=13, estimator=DecisionTreeRegressor())

# Features used in the IterativeImputer model
df_missing = df[['default','education']]

# Model training and filling missing values
imputed_data = imputer.fit_transform(df_missing).astype(int)

# Update the filled values to the DataFrame
imputed_df = pd.DataFrame(imputed_data, columns=df_missing.columns)
```

3) Outlier Removal

We checked for outliers for all the variables with numeric values using the box plot. We used IQR to select the outliers of ‘cons.conf.idx’, setting the lower and upper bounds respectively as “Q1-1.5*IQR” and “Q3+1.5*IQR”. We concluded that only the outliers for the ‘cons.conf.idx’ variable should be removed. This is because the number of outliers of the ‘cons.conf.idx’ variable was very small, only constituting about 1.07% from the dataset.




```
Number of outliers obtained using IQRs: 353
Size of the DataFrame :  (32950, 17)
Size of the DataFrame :  (32597, 17)
```

4) Encoding the categorical values

Lastly, we went through the training and validation when 'yes' is set to 1 and 'no' is set to 0. However in this case, the f1 score calculated for four different cases of 'poutcome' and 'month' resulted in 0.3 approximately for all cases. These four cases are represented in detail in the ‘Training & Validation’ stage. This suggests there might be an issue in the data preprocessing stage and we decided to approach it from a different perspective.

According to general conventions, 'yes' usually means "positive" or "present," while 'no' means "negative" or "absent." Thus, it is generally a good practice to convert 'yes' to 1 and 'no' to 0. However, depending on the characteristics of the data and the purpose of the analysis, this can be reversed. If 'no' is more significant, 'no' can be set to 1 and 'yes' to 0.

A scenario where 'yes' can be set to 0 and 'no' to 1 is when there is a significant data imbalance.

When there is a large disparity in the ratio of 'yes' to 'no' in the dataset, it can affect the model's performance. Hence, considering the class ratio while deciding the conversion method can be crucial. For example, it might be considered to set the more important class to 1 based on the class ratio.

In the 'train.csv' dataset, out of 32950 'y' values, 29238 are 'no' and 3712 are 'yes', which indicates a significant data imbalance with a ratio of approximately 1:7.8766 for 'yes' to 'no.'

Therefore, we will set 'yes' to 0 and 'no' to 1.

However, to follow the general conventions, in the prediction.csv file, ‘yes’ should be 1 and ‘no’ should be 0. Consequently, the value will be inverted during the process of converting to an integer at the end.

```
df["default"] = df["default"].replace({"yes": 0, "no": 1})
df["housing"] = df["housing"].replace({"yes": 0, "no": 1})
df["loan"] = df["loan"].replace({"yes": 0, "no": 1})
df["y"] = df["y"].replace({"yes": 0, "no": 1})
```

If the string variable has two values, it will be converted to 0 or 1 and if it has more than two, it will be preprocessed and analyzed using dummy values.

```
var1 = "job"
one_hot1 = pd.get_dummies(df[var1], prefix=var1)
df = pd.concat([df, one_hot1], axis=1).drop([var1], axis=1)

var2 = "marital"
one_hot2 = pd.get_dummies(df[var2], prefix=var2)
df = pd.concat([df, one_hot2], axis=1).drop([var2], axis=1)
```

3. Training & Validation

First, we split the dataset to train set and test set. We set the random_state as 100, applied shuffling before splitting, and divided 25% of the dataset as a test set.

```
from sklearn.model_selection import train_test_split

random_state = 100
shuffle = True
test_size_ratio = 0.25

train_df, test_df = train_test_split(df, test_size=test_size_ratio, random_state=random_state, shuffle=shuffle)
print(train_df.shape, test_df.shape)
```

```
(24447, 38) (8150, 38)
```

Then, we divided each dataset by input variable and output variable. To get the input variable for the training dataset, we dropped the target variable in the column dimension. To get the output variable, we put the target variable. Later, we applied the same process in the test dataset. For better visualization and interpretation, we added target_names and feature_names.

```
#get input variable for the training dataset(drop the target)
target = 'y'

X_train = train_df.drop(target, axis=1).values
y_train = train_df[target].values

X_test = test_df.drop(target, axis=1).values
y_test = test_df[target].values

#used for visualization and interpretation
target_names = ["no", "yes"]
feature_names = train_df.drop(target, axis=1).columns
```

Since we could not decide if we should drop the ‘poutcome’ and the ‘month’ variables, we tried training by preprocessing them into dummy values or dropping the variables into four cases.

Case 1. Drop both ‘poutcome’ and ‘month’ variables

```
Baseline: [0.76065974 0.73298377 0.75025166 0.77914839 0.75255013]
L2: [0.76065849 0.73297913 0.75025441 0.77914663 0.75254889]
L1: [0.76062725 0.73299498 0.7502387  0.77910797 0.75248992]
Polynomial: [0.75900901 0.73017866 0.74532135 0.77516419 0.75058951]
DecisionTree: [0.77354065 0.76640434 0.75856597 0.7761163  0.75853125]
{'Baseline': 0.7551187400169143, 'L2': 0.7551175118113232, 'L1': 0.7550917627438268, 'Polynomial': 0.7520525461030921, 'DecisionTree': 0.7666317013885537}
```

```
Accuracy: 0.8990046127700898
F1: 0.34177215189873417
ROC AUC: 0.7854906294233519
```

Case 2. Set ‘poutcome’ into dummy values and drop ‘month’ variable

```
Baseline: [0.76130454 0.73342365 0.75054399 0.77969496 0.75304102]
L2: [0.7613037  0.73342365 0.75054439 0.77969409 0.75304019]
L1: [0.76127871 0.73343448 0.75053731 0.77965059 0.75298744]
Polynomial: [0.75947178 0.73054046 0.74554492 0.77553897 0.75090556]
DecisionTree: [0.77441079 0.76688751 0.75858208 0.77223514 0.75738046]
{'Baseline': 0.7556016326312907, 'L2': 0.7556012027341926, 'L1': 0.7555777063806026, 'Polynomial': 0.7524003386608038, 'DecisionTree': 0.7658991953011915}
```

```
Accuracy: 0.8979121145909201
F1: 0.3032311516155758
ROC AUC: 0.7867751246392023
```

Case 3. Drop ‘poutcome’ and set ‘month’ variable into dummy values

```
Baseline: [0.76384165 0.73594544 0.75344725 0.7813527  0.75514408]
L2: [0.7638379  0.73594428 0.75344332 0.78135226 0.75514076]
L1: [0.76382374 0.73595123 0.753439   0.78130261 0.75508885]
{'Baseline': 0.7579462238199529, 'L2': 0.7579437039522394, 'L1': 0.7579210864681993}
```

```
Polynomial: [0.76960898 0.73979305 0.75531046 0.78262599 0.76031952]
DecisionTree: [0.777701   0.75465838 0.76442906 0.7781273  0.76222678]
{'Polynomial': 0.7615316006543981, 'DecisionTree': 0.7674285012121873}
```

```
Accuracy: 0.8988832240835154
F1: 0.9452153896744492
ROC AUC: 0.780922681193432
```

Case4. Set both ‘poutcome’ and ‘month’ variables into dummy values

```
Baseline: [0.76435273 0.73636058 0.75375962 0.78178416 0.75554028]
L2: [0.7643519  0.73635826 0.75375765 0.78178504 0.75554111]
L1: [0.76432733 0.73635517 0.75373172 0.7817411  0.75549792]
```

```
Accuracy: 0.8982762806506434
F1: 0.31198686371100165
ROC AUC: 0.7879134898607072
```

Comparing all four cases, Case 3, where we dropped ‘poutcome’ and set ‘month’ variables into dummy values, showed the best performance. Therefore, we dropped the ‘poutcome’ variable additionally and set the ‘month’ variable into dummy values.

```
df.drop("id", axis=1, inplace=True)
df.drop("contact", axis=1, inplace=True)
df.drop("day_of_week", axis=1, inplace=True)
df.drop("poutcome", axis=1, inplace=True)
df.info()
```

```
var1 = "job"
one_hot1 = pd.get_dummies(df[var1], prefix=var1)
df = pd.concat([df, one_hot1], axis=1).drop([var1], axis=1)

var2 = "marital"
one_hot2 = pd.get_dummies(df[var2], prefix=var2)
df = pd.concat([df, one_hot2], axis=1).drop([var2], axis=1)

var3 = "month"
one_hot3 = pd.get_dummies(df[var3], prefix=var3)
df = pd.concat([df, one_hot3], axis=1).drop([var3], axis=1)
```

From Case 3, we moved on to training and validation with K-fold Cross Validation for a single model. We first defined the model as logistic regression model. Then, we defined the number of folds, whether to shuffle the data before splitting into batches, and the random state. Since we set the shuffle as True, the random state controls the randomness of each fold. The scores from each iteration show the accuracy score. Each result for applying the regularization term L1 and L2 all showed the same result as not applying them.

```
model = LogisticRegression(solver="saga", max_iter=10000, penalty=None)
kf = KFold(n_splits=5, shuffle=True, random_state=random_state)
scores = cross_val_score(model, X_train, y_train, cv=kf)

print("Scores from each iteration:", scores)
print("Average score:", scores.mean())
```

```
Scores from each iteration: [0.90451143 0.89156383 0.88789964 0.91015783 0.90226629]
Average score: 0.8992798027010143
```

Then, we applied K-fold Cross Validation for multiple models. The result below shows the performance of the logistic regression model, polynomial model, and decision tree model. We used the ROC AUC score to compare the results. The approximate average score for the logistic regression model is 0.7579. Validating the scores for the polynomial model took a fair amount of time because it validates more cases based on the degree. The average score for the polynomial model is about 0.7615. Lastly, the average score for the decision tree model is about 0.7674. Consequently, the score for the decision tree model was the highest.

```
max_iter = 10000

#dictionary for each model
models = {
  "Baseline": LogisticRegression(solver="saga", max_iter=max_iter, penalty=None),
  "L2": LogisticRegression(solver="saga", max_iter=max_iter, penalty="l2", C=1.0),
  "L1": LogisticRegression(solver="saga", max_iter=max_iter, penalty="l1", C=1.0),
  "Polynomial": Pipeline([("poly_features", PolynomialFeatures(degree=2)),
                          ("softmax_reg", LogisticRegression(solver="saga", max_iter=max_iter, penalty=None))]),
  "DecisionTree": DecisionTreeClassifier(criterion="gini", max_depth=5, min_samples_split=2, min_impurity_decrease=0.0)
}

kf = KFold(n_splits=5, shuffle=True, random_state=random_state)

scores = {}

#iterate for each item, and get the cross validation score
for name, model in models.items():
    _scores = cross_val_score(model, X_train, y_train, cv=kf, scoring="roc_auc")
    print("{}: {}".format(name, _scores))
    scores[name] = _scores.mean()

print(scores)
```

```
Baseline: [0.76384165 0.73594544 0.75344725 0.7813527  0.75514408]
L2: [0.7638379  0.73594428 0.75344332 0.78135226 0.75514076]
L1: [0.76382374 0.73595123 0.753439   0.78130261 0.75508885]
{'Baseline': 0.7579462238199529, 'L2': 0.7579437039522394, 'L1': 0.7579210864681993}
```

```
Polynomial: [0.76960898 0.73979305 0.75531046 0.78262599 0.76031952]
DecisionTree: [0.777701   0.75465838 0.76442906 0.7781273  0.76222678]
{'Polynomial': 0.7615316006543981, 'DecisionTree': 0.7674285012121873}
```

We also applied k-fold cross validation using grid search to the decision tree model because it showed the highest performance as above.

```
from sklearn.model_selection import GridSearchCV

model = DecisionTreeClassifier(min_samples_split=2)

# Define the hyperparameters and their possible values
param_grid = {
    "criterion": ["gini", "entropy"],
    "max_depth": [5, 10, 20, None],
    "min_impurity_decrease": [0.0, 0.05, 0.1],
}

grid_search = GridSearchCV(model, param_grid, cv=kf, scoring="roc_auc")
grid_search.fit(X_train, y_train)

# Print the best parameters and the corresponding score
print("Best parameters: ", grid_search.best_params_)
print("Best CV score: {:.6f}".format(grid_search.best_score_))
```

```
Best parameters:  {'criterion': 'entropy', 'max_depth': 5, 'min_impurity_decrease': 0.0}
Best CV score: 0.779332
```

We also tried the training and validation using the 6 ensemble methods. Each result is shown below.

1) AdaBoost result

```
Best parameters:  {'estimator': DecisionTreeRegressor(max_depth=3), 'learning_rate': 1.0, 'n_estimators': 25}
Best CV score: -0.080556
```

2) Bagging result

```
Best parameters:  {'learning_rate': 0.1, 'max_depth': 6, 'n_estimators': 25}
Best CV score: -0.078180
```

3) Gradient Boosting result

```
Best parameters:  {'learning_rate': 0.1, 'max_depth': 6, 'n_estimators': 25}
Best CV score: -0.078180
```

4) LightGBM result

```
Best parameters:  {'enable_bundle': True, 'n_estimators': 25, 'reg_alpha': 0.1, 'reg_lambda': 0}
Best CV score: -0.077933
```

5) Random Forest result

```
Best parameters:  {'max_features': 'sqrt', 'n_estimators': 50}
Best CV score: -0.086258
```

6) XGBoost result

```
Best parameters:  {'n_estimators': 25, 'reg_alpha': 0.1, 'reg_lambda': 0.1}
Best CV score: -0.078368
```

4. Evaluation

After all the training and validation, we found that among all the models, which include logistic regression, polynomial, decision tree, and 6 ensemble methods, **LightGBM** method showed the best performance with the lowest MSE and the highest R^2 value. Below is the performance result of the ensemble models.

```
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

for _name, _model in models.items():
    y_pred = _model.predict(X_test)
    print("{:>17}: MSE={:.4f} | MAE={:.4f} | R^2={:.4f}".format(
        _name,
        mean_squared_error(y_test, y_pred),
        mean_absolute_error(y_test, y_pred),
        r2_score(y_test, y_pred)
        )
    )
```

```
    Decision Tree: MSE=0.0790 | MAE=0.1570 | R^2=0.2072
          Bagging: MSE=0.0872 | MAE=0.1636 | R^2=0.1255
    Random Forest: MSE=0.0855 | MAE=0.1547 | R^2=0.1422
         AdaBoost: MSE=0.0788 | MAE=0.1599 | R^2=0.2099
Gradient Boosting: MSE=0.0781 | MAE=0.1590 | R^2=0.2162
          XGBoost: MSE=0.0780 | MAE=0.1589 | R^2=0.2170
         LightGBM: MSE=0.0771 | MAE=0.1595 | R^2=0.2267
```

Then, we used the decision tree model to evaluate the probability of each class. The left side shows the probability of class 0 and the right shows the probability of class 1. We can conclude that the probability of class 1 is way higher than class 0. This indicates that the majority of people will not subscribe to the bank account.

```
from sklearn.metrics import accuracy_score, f1_score, roc_auc_score

model = DecisionTreeClassifier(criterion="entropy",
                               max_depth=5,
                               min_samples_split=2,
                               min_impurity_decrease=0.0)

model.fit(X_train, y_train)

y_prob = model.predict_proba(X_test)
print("Estimated probs:", y_prob[:10])

y_cls = model.predict(X_test)
print("Estimated classes:", y_cls[:10])
print()
```

```
Estimated probs: [[0.05258108 0.94741892]
 [0.05258108 0.94741892]
 [0.83928571 0.16071429]
 [0.02433234 0.97566766]
 [0.02433234 0.97566766]
 [0.05258108 0.94741892]
 [0.02433234 0.97566766]
 [0.05258108 0.94741892]
 [0.05258108 0.94741892]
 [0.05258108 0.94741892]]
Estimated classes: [1 1 0 1 1 1 1 1 1 1]
```

With our final model, LightGBM, we evaluated the accuracy, f1 score, and ROC AUC scores. For accuracy and f1 score, we classified the data points and for ROC AUC, we put the probability for y=1. The result is shown below. The f1 score was pretty high, followed by accuracy and ROC AUC scores.

```
# Accuracy, F1, and ROC AUC scores
print("Accuracy:", accuracy_score(y_test, y_cls))
print("F1:", f1_score(y_test, y_cls))
print("ROC AUC:", roc_auc_score(y_test, y_prob[:, 1]))
```

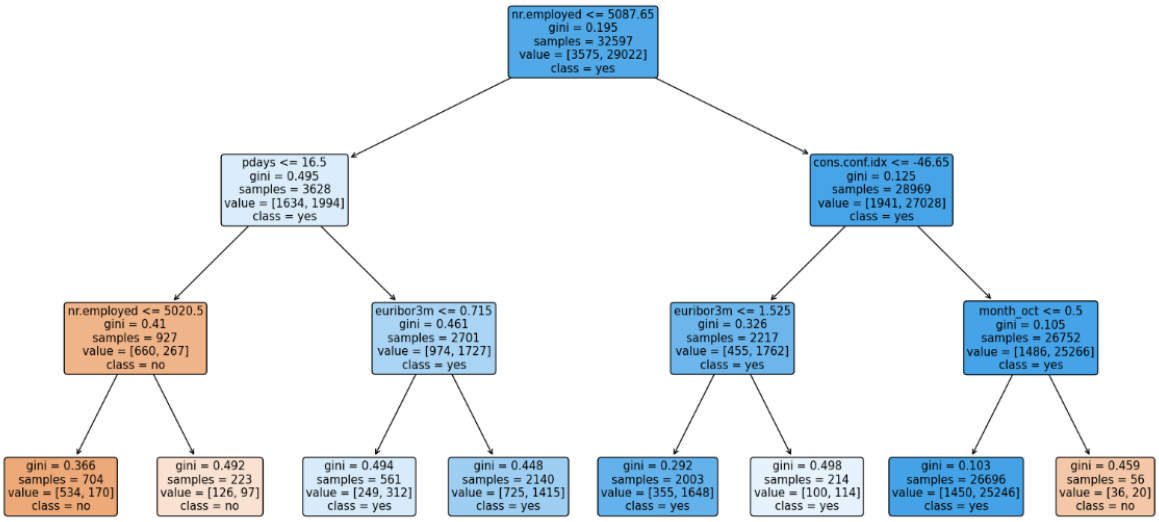
```
Accuracy: 0.9009815950920246
F1: 0.9464250149372635
ROC AUC: 0.7866876866578849
```

We used the confusion matrix to validate the numbers of actual counts and prediction counts. We had a total of 323 predicted 0s and 7827 predicted 1s. From here, the key value for our model is 7128, where actual 1 values and predicted 1 values match. As in the same context as decision tree classifier evaluation, we can conclude that a large portion of people will not subscribe to the bank account.

```
from sklearn.metrics import confusion_matrix

conf_matrix = confusion_matrix(y_test, y_cls)
conf_matrix_df = pd.DataFrame(
    conf_matrix,
    columns=["Predicted 0", "Predicted 1"],
    index=["Actual 0", "Actual 1"]
)
print(conf_matrix_df)
```

```
          Predicted 0  Predicted 1
Actual 0          215          699
Actual 1          108         7128
```

5. Interpretation



-Root Node: The data is split into two groups based on whether nr.employed is less than or equal to 5087.65.

-Left Subtree: Further splits are made based on whether pdays is less than or equal to 16.5 and additional conditions.

-Right Subtree: Further splits are made based on whether cons.conf.idx is less than or equal to -46.65 and additional conditions.

-In most nodes, the class is yes, indicating that the model classifies the majority of the data as positive.

-The Gini impurity indicates the impurity of each node; lower values signify purer nodes.

1. **Decision Tree Model**

-Root Node Split: The root node split on 'nr.employed' indicates that employment numbers significantly influence the likelihood of a term deposit subscription. Lower employment rates suggest a higher probability of subscription, possibly reflecting economic conditions where individuals seek secure savings options.

-Subsequent Splits: The further splits based on 'pdays' and 'cons.conf.idx' highlight the importance of recent marketing contact and consumer confidence. A lower number of days since last contact ('pdays') and a lower consumer confidence index ('cons.conf.idx') both correlate with a higher likelihood of subscription.

2. Model Performance Across Different Algorithms

-Logistic Regression: Provided a solid baseline with consistent performance, indicating linear relationships within the data. However, it was outperformed by more complex models.

-Polynomial and Decision Tree Models: These models captured more intricate patterns in the data. The decision tree, in particular, excelled due to its ability to handle non-linear relationships and interactions between features.

-Ensemble Methods: The superior performance of LightGBM demonstrates the power of boosting techniques in improving predictive accuracy and handling data imbalance. LightGBM's robustness and efficiency make it suitable for large datasets and complex feature interactions.

3. Ensemble Model Insights

-LightGBM Performance: LightGBM's high F1 score, combined with its low MSE and high R^2 values, underscores its effectiveness in balancing precision and recall. This makes it particularly adept at identifying both subscribers and non-subscribers accurately.

-Feature Importance: Analysis of feature importance from LightGBM reveals that economic indicators ('nr.employed', 'emp.var.rate') and recent marketing efforts ('pdays') are critical predictors. This insight can guide future marketing strategies and economic assessments.

4. Confusion Matrix Analysis

-The confusion matrix highlights the model's reliability in predicting non-subscriptions, with a high number of true negatives ('no' predictions). This indicates that the model effectively identifies individuals unlikely to subscribe, which can be valuable for targeted marketing efforts to focus on potential subscribers.

We once again note that due to significant data imbalance, we have trained the model with 'yes' as 0 and 'no' as 1. However, to follow the general conventions, in the prediction.csv file, 'yes' is set to 1 and 'no' is set to 0. You can verify in the csv file.